PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, HIGH PERFORMANCE, BA-X093/U

This specification is approved for use within Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-49471(ER).

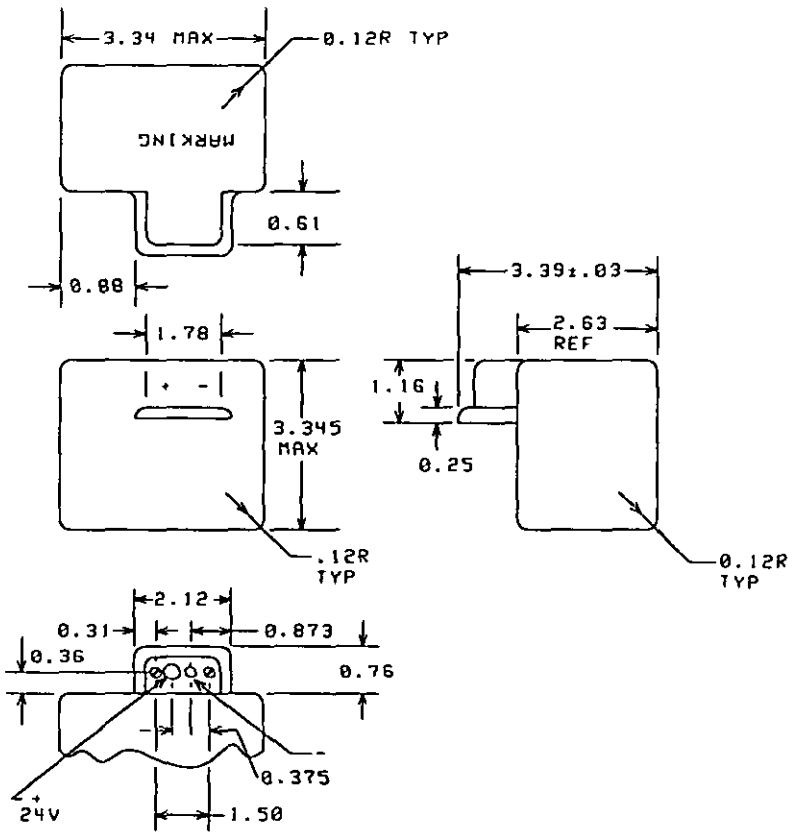


Figure 1. Battery dimensions.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:12Z
Check the source to verify that this is the current version before use.

NOTES:

1. All dimensions are in inches.

2. The battery connector shall be a flat, fiberglass socket connector with two (2) push-in terminals per MIL-B-18 as shown in Figure 1.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

Maximum voltage: 27.45 volts.

Minimum voltage (for capacity): 18.00 volts.

Safety features: Applicable.

Charge protection: Applicable.

High temperature device: Applicable.

Complete discharge device: Applicable.

Terminals: Two (2) hole flat plate type (with lead-in countersink on both holes) female socket per MIL-B-18, Type IV, with no obstruction of any holes.

Weight (maximum): 1.43 pounds (650 grams).

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Minimum Capacity Requirements in Hours to 18.00 Volts
I	3.4
L	1.7
H	6.8
HT	5.5
LT	1.6
IT	3.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:12Z
Check the source to verify that this is the current version before use.

Initial voltage delay: During the I, H, HT, and IT discharge tests, initial voltages below the 18.00 volt minimum voltage cannot exceed a 5.0 second duration. During L and LT discharge tests, initial voltage below 18.00 volts cannot exceed a 60.0 second duration.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage:

Details on storage conditions for all capacity tests are described in the basic specification.

(2) Discharge:

Cells (forced discharge): Two cells shall be discharged at 1.0 ampere at 70±5°F to 2/3 of open-circuit voltage. Each discharged cell shall then be connected in series to eight fresh cells. Each string shall be discharged at 1.0 ampere at 70±5°F to 2/3 open-circuit voltage of the eight fresh cells.

Batteries: For all capacity tests, except the H and HT, the batteries shall be discharged at 1.0 ampere constant current to zero volts. For the H and HT capacity tests, the batteries shall be discharged through 1.0 ampere constant current for five minutes followed by open-circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts.

Closed-circuit voltage: Load resistance of 48 ohms shall be used between the battery terminals. Voltage shall be 18.0 volts or greater within five seconds.

Custodian:
Army - ER

Preparing Activity:
Army - ER
(Project 6135-A274-12)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:12Z
Check the source to verify that this is the current version before use.